

HSF Lab Notebook Entry: Emergent $SU(3)$ and Gauss Invariance

Objective

Test whether the emergent $\mathfrak{su}(3)$ algebra obtained from qutrit echo dynamics can produce exact Gauss invariance under the strict *no-refolding* constraint.

Targets:

- Confirm full $\mathfrak{su}(3)$ closure when $d_{\text{site}} = d_B = 3$.
- Construct explicit bidirectional link generators L^a, R^a from echo algebra extraction.
- Build a gauge-covariant Hamiltonian

$$H = \sum_a (Q_x^a \otimes L^a \otimes I_y + I_x \otimes R^a \otimes Q_y^a)$$

- Test Gauss commutators for an oriented link $x \rightarrow y$:

$$G_x^a = Q_x^a + L^a, \quad G_y^a = Q_y^a - R^a$$

- Determine whether $[H, G_x^a] \approx 0$ and $[H, G_y^a] \approx 0$ (numerical tolerance).

Key Context from Results

Emergent $SU(3)$ Strengthening Result

From qutrit-site runs:

- Every row: `basis_dim` = 8.
- Singular values: all large; no ε -scale stragglers.
- Commutator closure: perfect across all 28 generator pairs.
- Held for aligned and mixed coupling models; held across ε values.

Comparison baseline (qubit-site with $d_B = 3$):

- `basis_dim` topped out at 6–7.
- Several singular values were ε -scale.

Conclusion:

$\mathfrak{su}(d_B)$ emerges fully when $d_{\text{site}} = d_B$.

Interpretation: matching matter degrees of freedom and mediator (bond-register) degrees of freedom is required for full gauge-sized algebra emergence (e.g., $SU(3)$ color requires triplet matter).

No-Unfolding: Strict Form (Version A)

Rationale adopted: if attachment/boundary sectors can be rewritten after matter exists, then structural information is destroyed (no-forgetting violation) and influence routing can be altered in a way that threatens no-signaling.

Therefore, strict no-unfolding implies:

$$[H, L^a] = 0, \quad [H, R^a] = 0$$

as an attachment-preservation/superselection requirement.

Scripts Used and What They Established

Echo Algebra and L/R Basis Extraction

- `echo_algebra_step1_qutrit_sites_su3.py`
- Output artifacts used in later tests:
 - `echo_algebra_step1_qutrit_su3_LR_bases_20260219_144601.npz`
 - `echo_algebra_step1_qutrit_su3_LR_20260219_144601.json`

Established: extracted L^a and R^a on a qutrit bond register form a robust 8D algebra with perfect closure consistent with $\mathfrak{su}(3)$.

Strict No-Unfolding Attachment Checker (Sanity + Diagnostics)

- `hsf_no_unfolding_attachment_checker_v1.py`
- `hsf_no_unfolding_attachment_checker_v2.py` (rewritten to fix output formatting; ran successfully)

Data:

- `hsf_no_unfolding_attachment_checker_v1_20260221_105825.json`
- `hsf_no_unfolding_attachment_checker_v2_20260221_124732.json`

Observation: using the *echo* Hamiltonian in the checker gives Gauss-commutator magnitudes $\mathcal{O}(1)$ and (for the symmetric echo ansatz) often uniform across generators (a normalization signature), indicating the echo Hamiltonian is not gauge-covariant by construction.

Gauge-Covariant Hamiltonian Minimal Test

- `gauss_covariant_hamiltonian_su3_minimal_v1.py`

Data:

- `gauss_covariant_hamiltonian_su3_minimal_v1_20260221_125741.json`

Tested explicit gauge-covariant contraction:

$$H = \sum_a (Q_x^a \otimes L^a \otimes I_y + I_x \otimes R^a \otimes Q_y^a).$$

Result: Gauss commutators remain $\mathcal{O}(1)$ (fails tolerance), meaning the failure is not just because the echo ansatz was the “wrong tensor structure.”

Adjoint Alignment Attempt

- `gauss_alignment_su3_v1.py`

Data:

- `gauss_alignment_su3_v1_20260221_162513.json`

Attempt: align link algebras to the site algebra by solving an orthogonal alignment problem on adjoint matrices (derived from structure constants).

Result:

- Alignment objectives remained $\mathcal{O}(10)$ (did not approach ≈ 0).
- Gauss commutators did not drop to numerical noise; still $\mathcal{O}(1)$.

Conclusion: not a simple basis relabeling.

Structure-Constants / Representation Variant Diagnostic

- `gauss_structure_constants_diagnostic_su3_v1.py`

Data:

- `gauss_structure_constants_diagnostic_su3_v1_20260221_170108.json`

Tested all 16 combinations of representation conventions for (L, R) :

$$\text{orig}, -T, T^T, -T^T$$

(including the conjugate-representation convention $-T^T$).

Result:

- No variant combination produced Gauss invariance (no pass at tol).
- Alignment objectives stayed $\mathcal{O}(10)$ rather than collapsing.

Conclusion:

Not a basis mismatch, not a sign mismatch, not a conjugate-rep convention issue.

Instead:

Structure constants for the extracted link algebra are not orthogonally equivalent to the site algebra.

Final Conclusions from This Session

1. Confirmed computational rule:

$\mathfrak{su}(d_B)$ emerges fully when $d_{\text{site}} = d_B$.

Qutrit sites ($d_{\text{site}} = 3$) produce a robust, full 8D algebra with no weak directions.

2. **Strict no-refolding interpreted as superselection/attachment preservation:** once matter exists, attachment rewriting is forbidden (otherwise it implies information loss and threatens locality constraints).
3. **Gauss invariance is not recovered by simply using extracted $\mathfrak{su}(3)$ generators:** even the explicitly gauge-covariant Hamiltonian contraction fails Gauss at $\mathcal{O}(1)$.
4. **The obstruction is structural:** the extracted link $\mathfrak{su}(3)$ is a valid Lie algebra with closure, but its structure constants are not orthogonally equivalent to the canonical site $\mathfrak{su}(3)$ (not fixable by basis rotation, sign flip, transpose, or conjugate-rep convention).
5. **This identifies a real fork:** “Echo $SU(3)$ ” (Lie algebra emergence) is confirmed, but “Gauge $SU(3)$ ” (Gauss-invariant local gauge theory with fundamental triplet matter) is not yet obtained.

Next Directions (Fork)

Three branches to explore next:

1. **Change the matter representation:** the emergent matter sector might not be fundamental triplets compatible with the extracted link action.
2. **Derive a representation-matching constraint:** identify which additional constraint (bandwidth, stability, no-forgetting topology, etc.) forces the link algebra embedding to match matter.
3. **Confining/loop phase interpretation:** accept that the model may naturally yield an $SU(3)$ loop/interaction algebra without local Gauss symmetry.